

Eval Report of Insurance claims pre-processing agent

Run: 02 | September 2026

Backend model(s): openai/gpt-oss-20b

Scenario set: 12 scenarios × 5 runs = 60 triage cells

Executive summary

Within the scope of this evaluation, the agent's reasoning is strong enough to pre-screen claims, routing each document to one of four decisions (APPROVE, FLAG_FOR_REVIEW, DENY, or NEEDS_INFO) for a human reviewer: it never approved a claim it should have escalated, and it reached the correct decision on every run. It is not ready for deployment yet. A technical defect must be fixed first: when the provider rate-limits the structured-output call that converts reasoning into a typed decision, the agent falls back to a text parser that can misread an otherwise correct decision and deliver the wrong one.

Because the two ways to be wrong carry very different costs, the decision-critical numbers are the directional error rates, not a single accuracy figure. The agent approved nothing that should have been escalated or denied (false-approval rate 0%), and it wrongly denied one case that should have gone to a human (false-denial rate 2%). It routed every fraud and risk case to a human, correctly denied the out-of-scope claims, and asked for missing information whenever a required field was absent. The one delivered miss, a depressed confidence signal, and the structural-expectation shortfall all trace to the same cause: the provider rate-limiting this sweep, not the agent's reasoning.

Metric	Result	Grade
False-approval rate (missed escalation)	0% (0 of 45 non-approve-eligible runs)	A
False-denial rate (wrongful rejection)	2% (1 of 45 non-deny-eligible runs)	A
Guardrail violations (unverified-policy approvals)	0 / 60	A
Decision stability (agreement across repeats)	98%	A
Reasoning accuracy (correct conclusion reached)	100% (60/60)	A
Delivered-decision accuracy	98% (95% CI: 91% to 99.7%)	A
Structural expectations met	92% (37/40 scored runs)	B
Completion rate	100% (60/60, zero hard errors)	A
Mean latency	30.1s	-
Mean cost per triage	\$0.00108	-

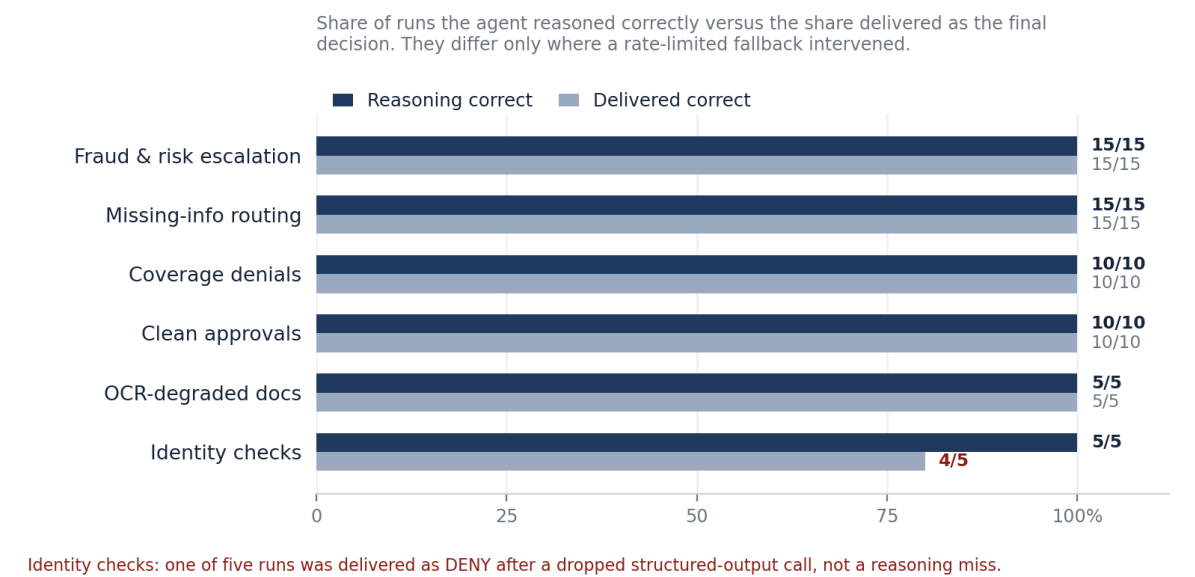
Score by risk category

This breakdown measures reasoning correctness: whether the agent reached an acceptable decision for each scenario type. It did so in all six categories, on every run. The identity-check row scores 5/5

on reasoning: the agent caught the name mismatch in all five runs. One run's typed decision was still corrupted downstream (see The one miss), and that delivered miss is counted separately in delivered-decision accuracy and the confusion matrix.

EXHIBIT 1 · REASONING VERSUS DELIVERED, BY CATEGORY

Reasoning is correct in every category



Reasoning versus delivered decisions, by risk category

Category	Scenarios	Reasoning correct
Clean approvals	clean_approve, confirmation_resolves	10/10
Coverage denials	expired_policy, non_claim	10/10
Missing-info routing	missing_policy, unverifiable_policy, prompt_injection	15/15
Fraud & risk escalation	fraud_new_customer, near_limit, velocity_repeat	15/15
OCR-degraded documents	ocr_recovery	5/5
Identity/consistency checks	name_mismatch	5/5

All risk categories were recognised and assigned the correct decision on every run: all fifteen fraud and risk runs were flagged for review, the coverage denials were denied, and the identity and missing-info cases were routed for confirmation rather than approved.

Confusion matrix

EXHIBIT 2 • CONFUSION MATRIX (60 RUNS)

Fifty-nine of sixty runs on the diagonal

Rows are the expected decision, columns what the agent returned. The diagonal is correct.

		DECISION RETURNED			
		APPROVE	FLAG	DENY	NEEDS INFO
EXPECTED	APPROVE	15	0	0	0
	FLAG	0	19	1	0
	DENY	0	0	10	0
	NEEDS_INFO	0	0	0	15

The only error (a FLAG scored as DENY) is a rate-limited fallback misparse, not a reasoning miss.

Confusion matrix

This matrix shows the decisions the pipeline actually delivered. Every cell lands on the diagonal except one, and nothing that required escalation was approved. The single off-diagonal cell (red) is a FLAG case returned as DENY, examined in the next section.

The incorrect output decision

One of the five name_mismatch runs is the single delivered error, and it is not a reasoning failure. The claim lists Tom Brown against a policy held by Alice Brown, and the agent's written reasoning opens with "Triage Decision: FLAG_FOR_REVIEW" and correctly names the mismatch. But the structured-output call that turns that reasoning into a typed decision was rate-limited, so the agent fell back to _fallback_result, which recovers a decision by scanning the reasoning text for the first decision keyword and checks DENY before FLAG_FOR_REVIEW. This run used the word "deny" elsewhere in its reasoning, so the scan returned DENY at 0.30 confidence with a structured_output_unavailable flag. The verdict was right; the fix belongs in the fallback parser.

Confidence and provider reliability

Mean reported confidence in this sweep is 0.71, but that number is dragged down by thirteen runs pinned at exactly 0.30.

EXHIBIT 3 • CONFIDENCE OF EACH RUN

The low-confidence runs are rate-limited fallbacks



Confidence of each run

Those thirteen are the heuristic fallback: when the structured-output call is rate-limited mid-run, the agent returns a usable decision at a fixed 0.30 confidence with a `structured_output_unavailable` flag. The remaining forty-seven runs, where structured output succeeded, sit between 0.50 and 1.00 at a mean of 0.82. The fallbacks are also why the structural-expectation checks show three misses, all three fallback runs unable to populate the missing-field list. This is a run condition, not a property of the agent: the sweep was gathered against a saturated daily token cap. A re-run under normal capacity should show far fewer fallbacks and a cleaner confidence and structural read.

Recommendations

- 1 **Retry the rate-limited call.** The current build treats a rate-limited structured-output call as final and drops straight to the text parser. It should retry that call before falling back, so a throttled request during synthesis does not decide the outcome.
- 2 **Fix the fallback decision parser.** `_fallback_result` returns the first decision keyword it finds in the reasoning text and ranks DENY above FLAG_FOR_REVIEW, so a stray mention of "deny" can override a stated FLAG_FOR_REVIEW verdict. Read the decision the agent actually stated instead of the first keyword in the body, and break ties toward FLAG_FOR_REVIEW.

Methodology notes and limits

- Single model (openai/gpt-oss-20b), single provider. No cross-model comparison.
- 12 scenarios with one document each. The Wilson confidence interval is wide, and accuracy should be read as a point estimate with real uncertainty.
- The 60 runs are 5 repeats of 12 scenarios, not 60 independent trials. Within-scenario correlation puts the effective sample size closer to 21 observations.
- This evaluates internal consistency against a seeded database, not agreement with a real human adjuster. It's primarily a regression check on agent behavior.
- Source data: `eval/results/runs.jsonl`, scored by `eval/metrics.py`. System under test: `agents/triage_agent.py` on openai/gpt-oss-20b.